

TOMAS ESPAÑA

te6653@princeton.edu | GitHub | Website

EXPERIENCE

Scuola Normale Superiore of Pisa

Pisa, Italy

Research Intern

4 months – 2025

- Reinforcement Learning for Optimal Execution with the Queue Reactive Model
- Supervisors: Yadh Hafsi, Edoardo Vittori, Fabrizio Lillo

EconophysiX - CFM Chair of Econophysics & Complex Systems at Ecole Polytechnique

Paris, France

Research Intern

6 months – 2024

- High-dimensional Statistics, Random Matrix Theory, Portfolio Optimization
- Supervisors: Victor Le Coz, Matteo Smerlak, Jean-Philippe Bouchaud

CFM - Capital Fund Management

Paris, France

Quantitative Research Intern

6 months – 2023

- Market Latency (Queueing Theory & Optimization) and Market Microstructure for the Execution Strategies team
- Coded a latency simulator of the CFM order pipeline to improve execution speed.

EDUCATION

Operations Research & Financial Engineering - Princeton University

Princeton, NJ

First-year PhD student

2025 – Present

M2MO - Université Paris Cité

Paris, France

MS in Mathematics: Probability, Finance & Data Science

2024 – 2025

- One of the top-ranking MS in Probability and Finance (ex DEA Laure Elie). Double degree with CentraleSupélec.

CentraleSupélec - Université Paris-Saclay

Paris, France

BS and MS of Science and Engineering - French Engineering "Grande Ecole"

2021 – 2025

- Applied Mathematics

LEADERSHIP

President of the Beta Sigma Club - Paris

2023 – 2024

European Quant Finance Student Club

- Led a 10-member team to organize lectures with distinguished quants and workshops with top-tier market makers.

President of the Mountain Bike Club - CentraleSupélec

2022 – 2023

SKILLS & ACTIVITIES

- **Languages:** English (Fluent), French (Native), Spanish (Native)
- **Programming:** Daily: Python. Occasionally: Matlab, R.
- **Activities:** Road Cycling & Mountain Bike (7 years - Regional level), Soccer (7 years - Top regional championship and European tournaments)

PUBLICATIONS

Pre-prints:

- P. Bousseyroux, T. Espana, M. Smerlak. *Another Marčenko-Pastur for Kendall's Tau*. arXiv preprint, 2025
- T. Espana, V. Le Coz, M. Smerlak. *Kendall Correlation Coefficients for Portfolio Optimization*. arXiv preprint, 2024

CONFERENCES

- SIAM Conference on Financial Mathematics and Engineering, Miami, July 2025.
- Poster Presentation. 18th Financial Risks International Forum, Paris, March 2025.
- Poster Presentation. Brunel-Bielefeld Workshop on Random Matrix Theory and Applications, London, December 2024.
- Poster Presentation. Random Tensors and Related Topics, Paris, October 2024.